METRIC

MIL-W-70342
4 November 1988

MILITARY SPECIFICATION

WASHER, FLAT, METRIC
GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers flat metric washers.

1.2 Types. Washers shall be of the following types:

Type I	Regular
Type II	Narrow
Type III	Wide

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

QQ-P-35 - Passivation Treatments for Corrosion Resisting Steel
QQ-P-416 - Plating, Cadmium (Electrodeposited)
PPP-H-1581 - Hardware (Fasteners and Related Items), Packaging of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Commander, US Army Armament Research, Development and Engineering Center, ATTN: SMCAR-ESC-S, Picatinny Arsenal, NJ 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5310

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:25Z
Check the source to verify that this is the current version before use.

MILITARY

MIL-W-70342/1 - Washer, Flat, Regular Series (Type I), Carbon Steel, Metric
MIL-W-70342/2 - Washer, Flat, Regular Series (Type I), CRES, Metric
MIL-W-70342/3 - Washer, Flat, Narrow Series (Type II), Carbon Steel, Metric
MIL-W-70342/4 - Washer, Flat, Wide Series (Type III), High Strength, Alloy Steel, Metric
MIL-W-70342/5 - Washer, Flat, Wide Series (Type III), Carbon Steel, Metric

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1312 - Fasteners, Test Methods

(Copies of specifications, standards and other government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

ASTM

ASTM A109 - Steel, Strip, Carbon, Cold-Rolled

ASTM A453 - Bolting Materials, High-Temperature, 50 to 120 KSI Yield Strength, with Expansion Coefficients Comparable to Austenitic Sheets

ASTM A582 - Free-Machining, Stainless and Heat-Resisting Steel Bars, Hot-Rolled or Cold Finish

ASTM B633 - Electro-Deposited Coatings of Zinc on Iron and Steel

ASTM E18 - Rockwell Hardness and Rockwell Hardness of Metallic Materials, Standard Test Methods for

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:25Z
Check the source to verify that this is the current version before use.

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between requirements of this specification and the specification sheet, the latter shall govern (see 6.2).

3.2 Material. Recycled and reclaimed materials (see 3.1) shall be used to the maximum extent practicable.

3.2.1 Carbon steel. Carbon steel washers shall be manufactured from Grade 1008 (UNS G10080) thru Grade 1018 (UNS G10180), cold rolled, No. 4 Temper, No. 3 Finish in accordance with ASTM A109.

3.2.2 Alloy steel. Alloy steel washers shall be manufactured from material as specified in ASTM A453.

3.2.3 Corrosion-resistant steel (austenitic). Austenitic corrosion resistant steel washers shall conform to Type 303 (UNS S30300) as specified in ASTM A582.

3.3 Dimensions and tolerances. Dimensions and tolerances for all washers shall conform to the dimensional requirements of the applicable specification sheet or military standard.

3.3.1 Parallelism. Washer faces shall be parallel within 50 µm.

3.3.2 Flatness. Washer faces shall be flat within:

150 µm for hole sizes thru 13 mm
250 µm for hole sizes 14 mm thru 32 mm
380 µm for hole sizes over 32 mm

3.3.3 Concentricity. The outside diameter and the inside diameter shall be concentric within the limits as specified in Table I.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:25Z
Check the source to verify that this is the current version before use.

TABLE I. CONCENTRICITY

OUTSIDE DIAMETER (mm)		CONCENTRICITY (μm)
OVER	TO	
1	3	125
3	6	150
6	10	180
10	18	215
18	30	260
30	50	310
50	80	370

3.4 Performance characteristics. Washers shall meet the following performance requirements when subjected to the applicable test as specified (see 4.3).

3.5 Protective finish and surface treatment. The protective finish and surface treatment of the washer when specified in the applicable sheet or military standard shall be as follows (see 4.3.2).

3.5.1 Carbon or alloy steel washers. When zinc coating or cadmium plating is specified for carbon or alloy steel washers (see 3.1), they shall conform to 3.5.1.1 or 3.5.1.2, as applicable.

3.5.1.1 Zinc coating. Carbon or alloy steel washers shall be zinc coating in accordance with ASTM B633, Type II, Class FE/ZN8.

3.5.1.2 Cadmium plate. Carbon or alloy steel washers shall be cadmium plated in accordance with QQ-P-416, Type II, Class 2.

3.5.1.3 Cleaning, descaling, and passivation. Corrosion resistant steel washers shall be cleaned, descaled and passivated in accordance with QQ-P-35.

3.6 Hardness.

3.6.1 Carbon steel washers. Carbon steel washers shall have a maximum hardness of 65HRB.

3.6.2 Alloy steel washers. Alloy steel washers shall have a minimum hardness of 40HRC.

Source: https://assist.dla.mil -- Downloaded: 202640-23T03:25Z
Check the source to verify that this is the currentversion before use.

3.7 Cracks. Washers shall be free from cracks (see 6.4), flaws, and pits in any location.

3.8 Workmanship. Washers shall be flat and smooth and free from burrs, loose scales, sharp edges and all other defects that may affect their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Quality conformance inspection. Quality conformance inspections shall be as specified in Table II. Each washer that does not pass one or more of the tests shall be defective.

TABLE II. QUALITY CONFORMANCE INSPECTION

INSPECTION	REQUIREMENT PARAGRAPH	TEST PARAGRAPH
Group A		
Dimensions	3.3	4.3.1
Protective finish and surface treatment	3.5	4.3.2
Group C		
Hardness	3.6	4.3.3

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:25Z
Check the source to verify that this is the current version before use.

4.2.1 Inspection lot. An inspection lot shall consist of all washers covered by a single specification sheet or military standard, produced under essentially the same conditions, and offered for inspection at one time.

4.2.2 Sampling for Group A inspection. Sampling for Group A inspection shall be as specified in 4.2.2.1 and 4.2.2.2.

4.2.2.1 Sampling for dimensions. Sampling shall be in accordance with Inspection Level S-3 of MIL-STD-105. The Acceptable Quality Level (AQL) shall be as indicated in Table III. The AQL shall apply to each individual defect, not to a group of defects.

4.2.2.2 Sampling for protective finish and surface treatment. Sampling for protecive finish and surface treatment test of washers shall be in accordance with applicable specifications referenced in 3.5.1.1 thru 3.5.1.3.

TABLE III. CLASSIFICATION OF DEFECTS

CATEGORY	DEFECT	INSPECTION METHOD
Critical	None defined	
Major	AQL = 2.5 percent defective	
101	Diameters (see 3.3)	CIE 1/
102	Parallelism (see 3.3.1)	CIE 1/
103	Flatness (see 3.3.2)	CIE 1/
104	Concentricity (see 3.3.3)	CIE 1/
Minor	AQL = 4.0 percent defective	
201	Protective finish and surface treatment missing (see 3.5 thru 3.5.1.3)	Visual
202	Workmanship (see 3.8)	Visual

1/ Commercial inspection equipment

4.2.3 Sampling for Group C inspection. Sampling shall be in accordance with Level S-1 of MIL-STD-105. The AQL shall be 1.5 percent defective. Group C inspection shall be performed on sample units which have passed Group A inspection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:25Z
Check the source to verify that this is the current version before use.

4.2.4 Noncompliance. If a sample fails to pass Group C inspection, the manufacturer shall notify the qualifying activity and the cognizant inspection activity of such failure and take corrective action on the materials or processes, or both, as warranted, and on all units of product which can be corrected and which were manufactured with essentially the same materials and processes, and which are considered subject to the same failure. Acceptance and shipment of the product shall be discontinued until corrective action, acceptable to the qualifying activity has been taken. After the corrective action has been taken, Group C inspection shall be repeated on additional sample units (all tests and examinations, or the test which the original sample failed, at the option of the qualifying activity). Group A inspections may be reinstituted; however, final acceptance and shipment shall be withheld until the Group C inspection has shown that the corrective action was successful. In the event of failure after reinspection, information concerning the failure shall be furnished to the cognizant inspection activity and the qualifying activity.

4.3 Methods of inspection.

4.3.1 Visual and dimensional examination. Each washer taken as specified in 4.2.2.1 shall be examined to verify conformance with this specification. Examination shall be conducted in accordance with Table III.

4.3.2 Protective finish and surface treatment test. The carbon or alloy steel washers shall conform to the applicable protective finish and surface treatment test QQ-P-416 and ASTM B633 as specified in 3.5.1. Corrosion resistant steel washers shall meet requirements of 3.5.1.3.

4.3.3 Hardness test. Washers shall be tested for surface hardness in accordance with MIL-STD-1312, Test Method 6.

4.3.4 Materials inspection. Materials inspection shall consist of certification supported by verifying data that the materials used in fabricating the washers are in accordance with the requirements of 3.2 prior to such fabrication.

4.3.5 Inspection of packaging. The sampling and inspection of the preservation, packing, and container marking shall be in accordance with the requirements of PPP-H-1581.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with PPP-H-1581 (see 6.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:25Z
Check the source to verify that this is the current version before use.

6. NOTES

6.1 Intended use. This specification covers flat washers, for general purpose, aircraft and general metric bolt and nut application. Their purpose is to provide an increased bearing surface area, and to prevent galling of the base material.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification and applicable specification sheet or applicable Military Standard.

b. Applicable military specification number(s) or standard(s) (see 3.1).

c. Level (degree) of packaging protection in accordance with PPP-H-1581 (see 5.1).

6.3 Subject term (key word) listing.

Washer, Flat, Metric
Washer, Flat, Carbon Steel
Washer, Flat, Cres
Washer, Flat, Narrow Series
Washer, Flat, High Strength
Washer, Flat, Wide Series

6.4 Definition. A crack is a clean crystalline fracture passing through or across the grain boundaries without inclusion of foreign elements.

Custodians:
Army - AR
Navy - AS
Air Force - 99

Review activities:
Army - ER
DLA - IS

User activities:
Army - CE, GL, MI
Navy - MC, SH, YD

Preparing activity:
Army - AR

Agent:
DLA - IS

(Project 5310-1576)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:25Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS. In a continuing effort to make our standardization documents better, the DoD provides this form for use in mitting comments and suggestions for improvements All users of military standardization documents are invited to provide testions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and led In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems Enter in block 6 any remarks not related to a specific paragraph of the document If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE ARMY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO 12062 WASHINGTON D C

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

Commander
US Army ARDEC
ATTN: SMCAR-BAC-S
Picatinny Arsenal, NJ 07806-5000

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:25Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1 DOCUMENT NUMBER	2 DOCUMENT TITLE
MIL-W-70342	

3a. NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5 PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6 REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

Source: https://assist.dla.mil -- Downloaded: 2026-10-28T03:25Z
Check the source to verify that this is the current version before use.